GPM-GS: 3D Gaussian Splatting Guided by Geometric Probability Models for Surface Reconstruction

Supplementary Material

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I. Additional Results

A. All Reconstructed Meshes

This supplementary material accompanies **GPM-GS: 3D Gaussian Splatting Guided by Geometric Probability Models for Surface Reconstruction** and presents all reconstructed meshes on the DTU [1], Tanks and Temples (TnT) [2], MipNeRF360 [3], and Replica [4] datasets. For completeness, some content may overlap with the main paper.

We first use GPM-GS to train the Gaussian radiance field. All images are downsampled by a factor of 2. Following GeoSVR [5], the original MipNeRF360 data are downsampled by a factor of 2 or 4 to accommodate both indoor and outdoor scenes. For Replica, 200 frames are uniformly sampled from each scene for training. After rendering depth maps from the trained Gaussian radiance field, we extract meshes using the truncated signed distance function (TSDF) algorithm. The voxel sizes for DTU and Replica are set to 0.002 and 0.005, respectively. For TnT and MipNeRF360, predefined bounding boxes are used with a scene resolution of 2048.

Our basic experimental procedures and hyperparameters are consistent with previous works [6], [7]. Notably, the open-source model DepthPro [8] is adopted to generate monocular depth maps. Depth calibration is performed twice at the 3,000 and 8,000 training iterations, and loss items $\mathcal{L}_{unc}$, $\mathcal{L}_{calib}$ and $\mathcal{L}_{wpcc}$ are only utilized within the first 20,000 iterations. The weights of λ_{scale} and λ_{mv} are kept consistent with [7], while λ_{unc} is set to 0.01, $\lambda_{calib} \in [0.1,\ 0.5]$ and $\lambda_{wpcc} \in [0.2,\ 0.01]$ are subjected to exponential decay, respectively. The bandwidths for measuring significant differences are set to $\lambda_d = 0.05$, $\lambda_n = 5$ and $\lambda_u = 1$, respectively. The predefined visibility probability threshold in depth calibration is chosen as $\tau_v = 0.5$. Additionally, we determine the initial source image pool based on the sorted number of feature points from the original structure from motion (SfM), and start updating the view-selection weights only after at least one round of multi-view supervision has been performed.

Figs. 1 - 4 present all reconstruction results of GPM-GS on the DTU, TnT, MipNeRF360, and Replica datasets, respectively.

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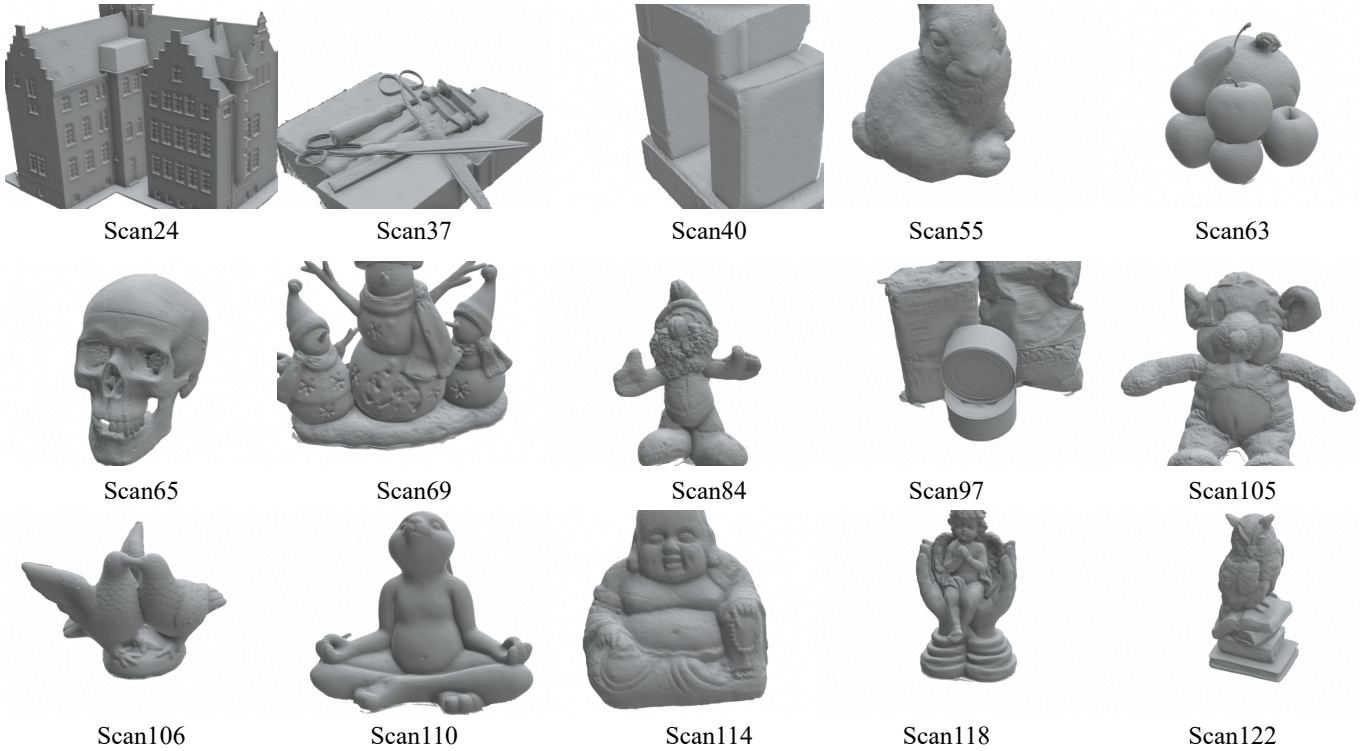

Fig. 1: Visualization of all reconstructed meshes on the DTU [1] dataset.

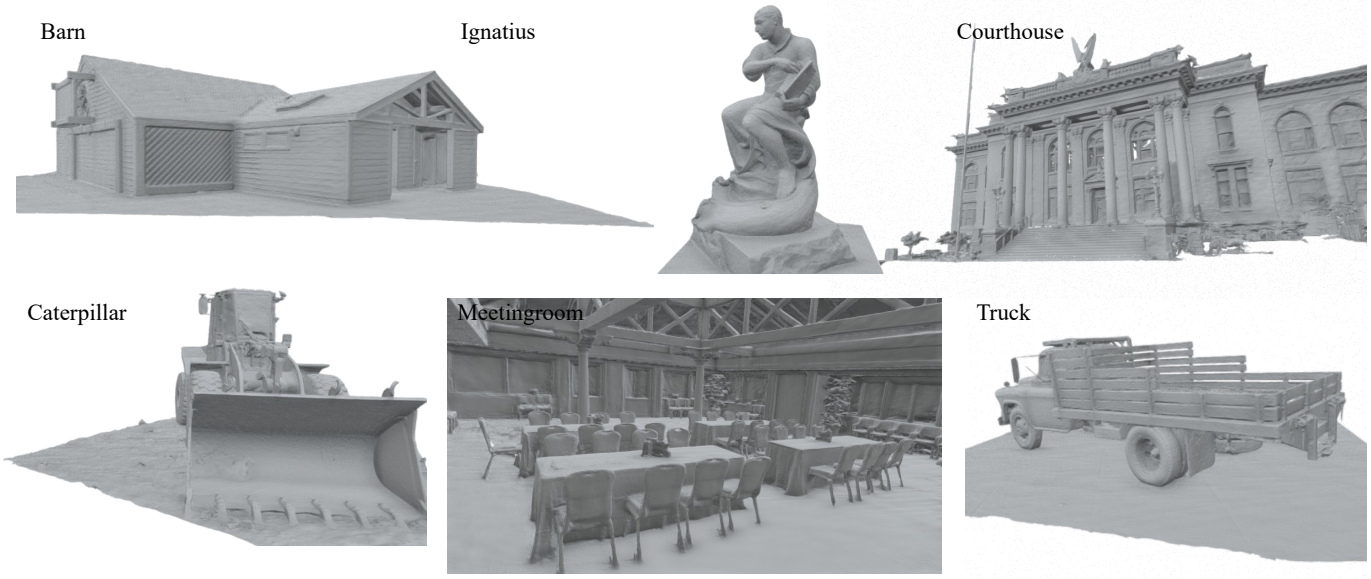

Fig. 2: Visualization of all reconstructed meshes on the TnT [2] dataset.

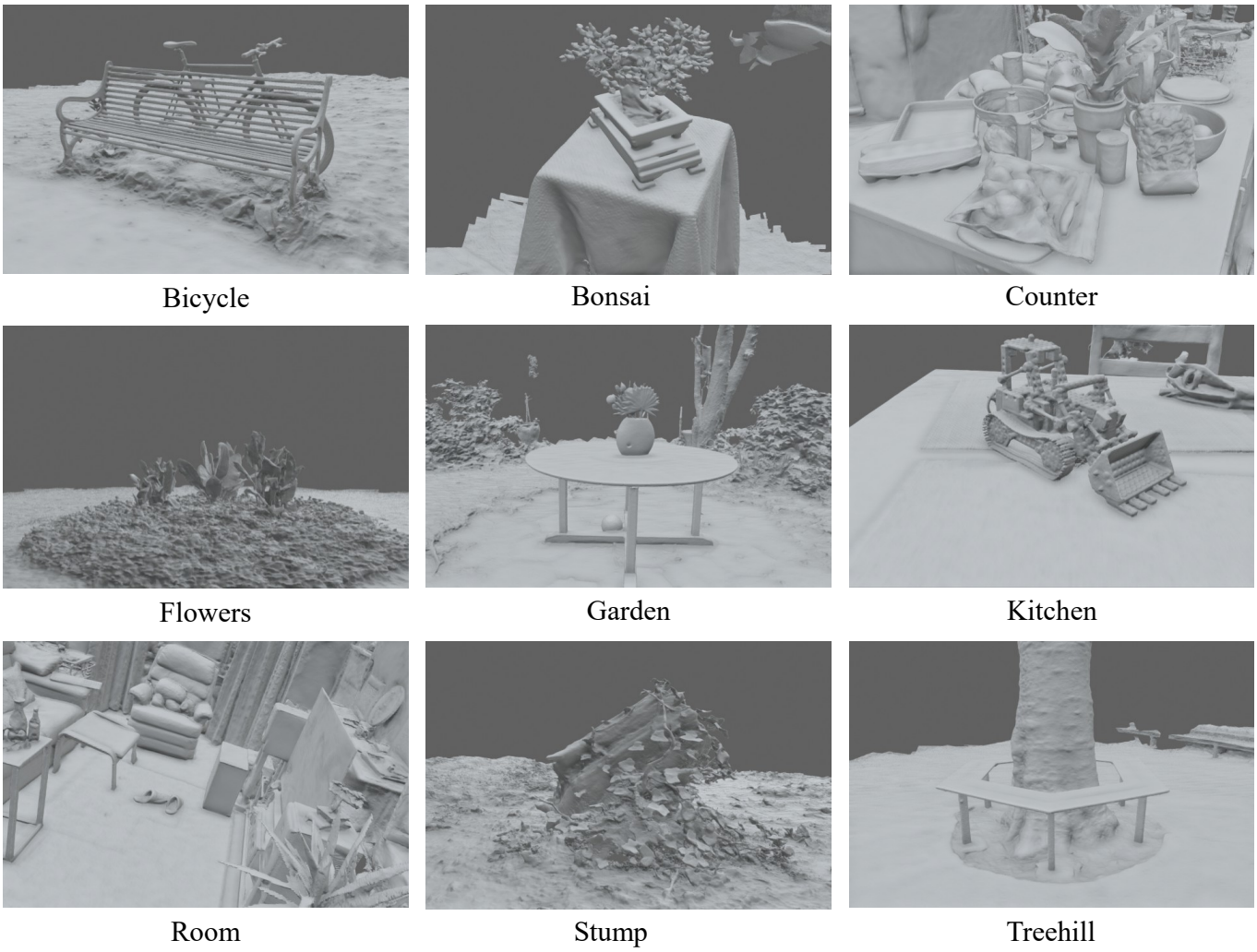

Fig. 3: Visualization of all reconstructed meshes on the MipNeRF360 [3] dataset.

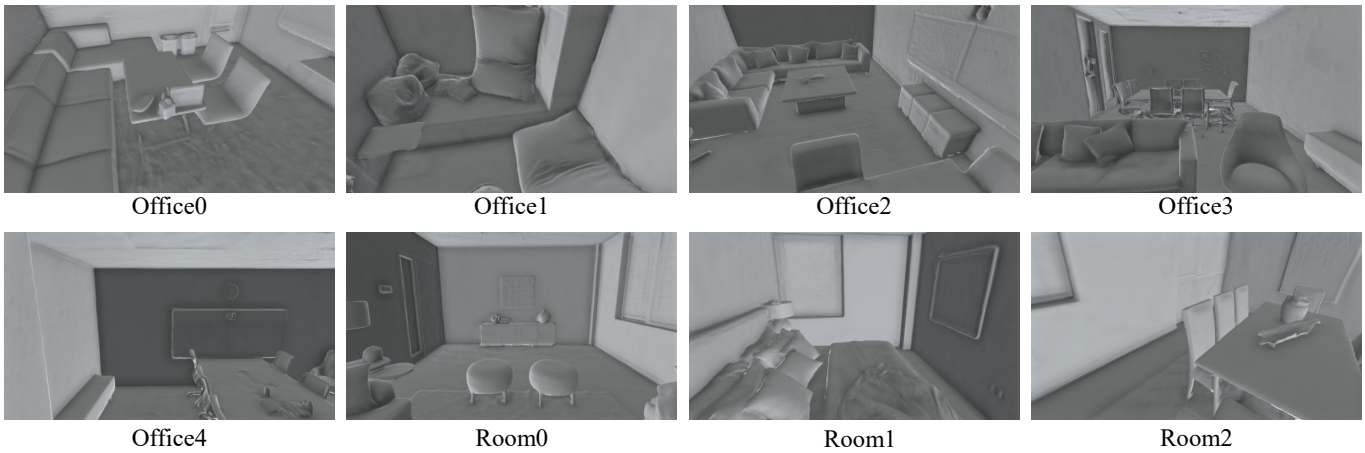

Fig. 4: Visualization of all reconstructed meshes on the Replica [4] dataset.